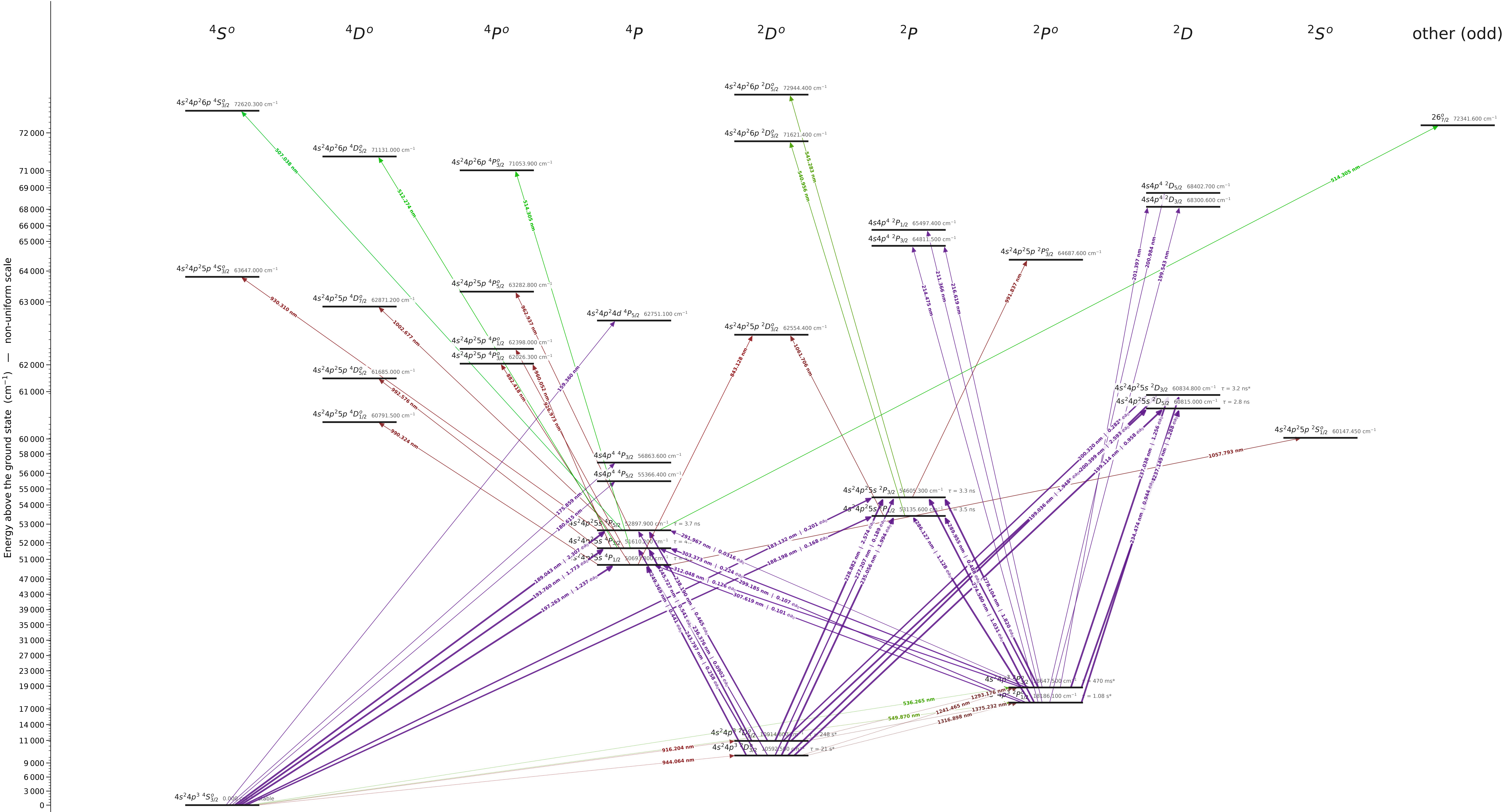
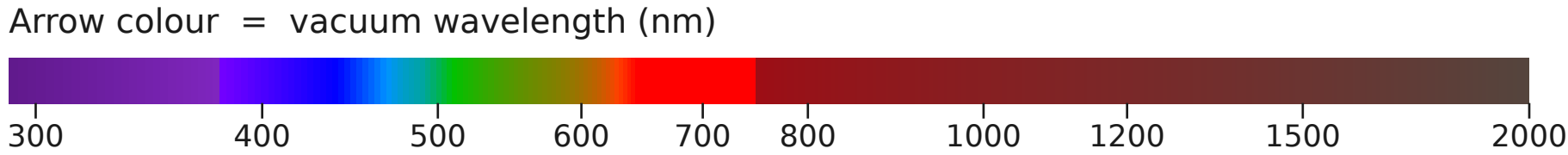


Arsenic

As (As I). Strongest classified transitions below 2 μm (NIST): 35 levels, 64 lines.

As ⁺ ionisation limit 78950.00 cm⁻¹ = 9.78855 eV (126.662 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
4s ² 4p ³ ⁴S _{3/2} – 4s ² 4p ² 5s ⁴P _{3/2}	193.7601	–	1547.2349	1.773	2.190e+08	37.01	94	resonance line
4s ² 4p ³ ⁴S _{3/2} – 4s ² 4p ² 5s ⁴P _{1/2}	197.2628	–	1519.7619	1.237	2.020e+08	35.37	94	resonance line
4s ² 4p ³ ⁴S _{3/2} – 4s ² 4p ³ ²P _{3/2}	536.2649	536.1158	559.0380	–	1.600e+00	3.386e-07	75.2	forbidden line within the ground configuration
4s ² 4p ³ ⁴S _{3/2} – 4s ² 4p ³ ²P _{1/2}	549.8705	549.7178	545.2056	–	6.910e-01	1.474e-07	74.6	forbidden line within the ground configuration
4s ² 4p ³ ⁴S _{3/2} – 4s ² 4p ³ ²D _{5/2}	916.2040	915.9526	327.2115	–	5.300e-03	6.418e-10	100	forbidden line within the ground configuration
4s ² 4p ³ ⁴S _{3/2} – 4s ² 4p ³ ²D _{3/2}	944.0642	943.8053	317.5552	–	7.490e-02	7.579e-09	100	forbidden line within the ground configuration
4s ² 4p ³ ²D _{3/2} – 4s ² 4p ³ ²P _{3/2}	1241.4649	1241.1253	241.4828	–	6.720e-01	3.386e-07	31.6	forbidden line within the ground configuration
4s ² 4p ³ ²D _{5/2} – 4s ² 4p ³ ²P _{3/2}	1293.1759	1292.8223	231.8265	–	4.400e-01	3.386e-07	20.7	forbidden line within the ground configuration
4s ² 4p ³ ²D _{3/2} – 4s ² 4p ³ ²P _{1/2}	1316.8984	1316.5384	227.6504	–	4.200e-01	1.474e-07	45.4	forbidden line within the ground configuration
4s ² 4p ³ ²D _{5/2} – 4s ² 4p ³ ²P _{1/2}	1375.2321	1374.8561	217.9941	–	5.900e-02	1.474e-07	6.37	forbidden line within the ground configuration

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁷⁵As hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
4s ² 4p ³ ⁴S _{3/2}	-66.2040(10)	-0.535(3)	F=0: +247.60 F=1: +181.93 F=2: +50.05 F=3: -149.09
4s ² 4p ³ ²D _{3/2} ^o	480(9)	–	F=0: -1800.00 F=1: -1320.00 F=2: -360.00 F=3: +1080.00
4s ² 4p ³ ²D _{5/2} ^o	725(26)	–	F=1: -3806.25 F=2: -2356.25 F=3: -181.25 F=4: +2718.75
4s ² 4p ³ ²P _{1/2} ^o	1909(14)	–	F=1: -2386.25 F=2: +1431.75
4s ² 4p ³ ²P _{3/2}	359(59)	–	F=0: -1346.25 F=1: -987.25 F=2: -269.25 F=3: +807.75
4s ² 4p ² 5s ⁴P _{1/2}	609(15)	–	F=1: -760.75 F=2: +456.45
4s ² 4p ² 5s ⁴P _{3/2}	303(18)	–	F=0: -1135.50 F=1: -832.70 F=2: -227.10 F=3: +681.30
4s ² 4p ² 5s ⁴P _{5/2}	734(12)	–	F=1: -3856.12 F=2: -2387.12 F=3: -183.62 F=4: +2754.38
4s ² 4p ² 5s ⁴P _{1/2}	102(27)	–	F=1: -127.38 F=2: +76.42
4s ² 4p ² 5s ²P _{3/2}	597(15)	–	F=0: -2237.25 F=1: -1640.65 F=2: -447.45 F=3: +1342.35
4s ² 4p ² 5s ²D _{5/2}	833(18)	–	F=1: -4375.35 F=2: -2708.55 F=3: -208.35 F=4: +3125.25
4s ² 4p ² 5s ²D _{3/2}	720(90)	–	F=0: -2698.12 F=1: -1978.62 F=2: -539.62 F=3: +1618.88

Measured A, B constants; sources in the data file.